



DealAura

A Software Systems Design Proposal & Product Requirements Document

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DEALAURA: A SOFTWARE SYSTEM DESIGN PROPOSAL

Business Case

Executive Summary

This proposal outlines the development of a software application designed to address the inefficiencies and losses within current sales teams and processes. The proposed software called “DealAura” is a cloud-based software application that features AI and BI capabilities. The system is cloud-based and can be deployed to either desktop or mobile applications, which are intended to function as a sales tool, and standalone or integrated into existing Customer Relationship Management (CRM) platforms. The core features and objectives are to leverage artificial intelligence (AI) and large language models (LLMs) to automate and streamline the manual sales tasks, such as searching, booking, planning, and other functions, thereby enabling sales force professionals to allocate more of their time, skill, and energy to strategic planning and relationship-building activities.

Problem/Opportunity Statement

A significant challenge in today’s modern business setting and marketplace is the highly manual and redundant clerical or office requirements placed on sales teams. A substantial part of a salesperson or team’s time is devoted to time-consuming, low-productive repetitive tasks, including manual prospecting or research, the writing of personalized outreach communication, and the careful creation of customized sales proposals. This working inefficiency not only reduces the business and sales strategy, their revenue-generating profits, and activities, but also presents a clear opportunity to enhance sales productivity through new technological tools. The development of a single comprehensive software solution that consolidates many of these sales functions, their data sources, and automates these time-intensive processes will provide a significant competitive advantage to any business or sales team that uses it.

Business and Sales Solution

The proposed software program “DealAura” is a cloud-based software application that can be deployed to desktop or mobile apps. This software program is designed to serve as a consolidated location and tool for salespersons and teams. It does address the known business and sales

challenges through its AI deep-learning LLMs. It can operate as a standalone or as integrated into other applications through a backend API. The program and source code are highly secure from hackers or bad actors. The program does offer basic boilerplate core features, with added features at a premium. It can do the following basic functions:

- **Personalized Sales Research and Canvas:** The AI system can provide deep-level research into prospects or canvas areas, or demographic locations to find sales candidates.
- **Personalized Pitch Generator:** The AI system will be able to draft personalized pitches or pitch calls to prospects. This includes automated phone calls and emails. This also includes creating pitch decks or prospect materials for prospects to send by regular mail or email.
- **Personalized Proposals and Contacts Generator:** The AI system will be able to provide prospects with proposals, accept applications, process applications, and sales claims. The AI will be able to change or customize the proposals or agreements, deal with pricing issues, address project scheduling, and timelines, and send out paperwork or requirements for any licensing or permits needed.
- **Dealing with Sales Complaints and issues:** The AI system will take calls and record all complaints about any sales matters for review. The system will track these complaints and seek to identify the causes or issues, and to help resolve them if possible.
- **Churn or Cancellation of Contract Threat Detector:** The AI system will be able to look for KPI and key indicators of negative feedback or issues that may be flagged for review, or for the cancellation process. The system will see to address the issue(s) before a person calls to cancel.

This program offers a comprehensive, viable solution for businesses, salespeople, and sales force teams. This system is designed to streamline the whole sales force process and workflow, which ensures efficiency, accuracy, and consistency while also reducing the business and salesforce overhead costs.

Business Benefits

The benefits of this new software application program and system are plentiful, with little downside. The core benefits, as mentioned previously, include:

- Providing salespeople and teams with more options, tools, freedom, and opportunities.

- Providing a program and system that is highly secure, dependable, affordable, and effective.
- Reduce any overhead costs to the business, salespeople, or sales teams.
- Increasing revenue or profitability and viability for the business and sales teams.

System Development

Waterfall

This software application will be using a traditional waterfall for the first part of its lifecycle. This is because the waterfall system is well-documented and for a linear structured process, with step phases to completion. This program will not use the waterfall past the design phase, because the model lacks the required flexibility or agility needed for adaptation to changes.

Agile

This program will be using the Agile method starting past the design phase, because the Agile method uses iterative and incremental phases to help address ongoing issues or changes that are needed. This agile method is good for its ability, agility, flexibility, and customer needs. It is also good for the need to deploy the program quickly to market, or make new releases, or fix bug issues.

Extreme Programming (XP)

The usage of Extreme Programming (XP) is a part of the Agile methodology. However, it will not be needed for this program case, unless otherwise specified. The reason is that XP is focused more on the customer's needs or wants, rather than what is required or needed technologically, and for budgeting purposes. Too many customers want to have too many added features, bells, and whistles, which require exhaustive testing and revisions, and releases, which are either not productive, or feasible, or reliable, or even able. Thus, while customer wants and needs are important, they also must be weighed by the judgments of the software developers and product managers to determine if such added features are beneficial.

Scrum

Scrum is another part of Agile methodology and is most often used in many software production teams. Scrum is a promising idea for a product that may have a slower rollout in phases, or many

quick releases, or if they have too many bugs in the system. Scrum deals with short and time-sensitive iterations called 'sprints.' The sprint leader may be the team lead or scrum master. This Scrum Agile method may also be used both by the Scrum Master and/or a Product Owner. Product Owners do not actually own the business or software; rather, they are only to look after the real owner's interests in the product and categorize or order the workflow accordingly. Smaller or start-up companies may have only one person wearing both hats.

Hybrid Method

This software application system development will be under a hybrid model or method. The first part will use the Waterfall method for documentation, requirements, and design. The rest of the life cycle will be using Agile methodology. The program will not need to use XP but may use Scrum if needed. The software developers and product managers make such determinations.

System Acquisition

There are six known common sources for acquiring software:

- 1. Information Technology Services (ITS):** This method involves contracting or outsourcing the product with an outside technology company to develop, host, and maintain the software system. This is best used by companies that do not have or wish to develop in-house, or are trying to save money, time, or liability by outsourcing it.
- 2. Packaged Software:** This is referred to as *off-the-shelf* or *ready-to-use* software applications. This method seeks to provide generic or basic boilerplate software that any customer could use. It is a promising idea for low-cost solutions that produce a large volume of copies for a large customer base. However, this is not ideal for customization or specification of the software or solution.
- 3. Enterprise Solutions:** This type is targeted at large companies that have adequate funding for large-scale, all-in-one software solutions, such as an ERP or CRM system. These products are designed for trying to manage the entire company and multiple departments streamlined into one integrated software solution. It is best for large companies based on need, who have the funding, and who also have the time for implementation of the system.
- 4. Cloud Computing:** This scheme delivers access to the software and services by the cloud method, which is an online service and server provider. Cloud solutions are growing in

popularity and usage, and are good for their ability to provide availability, scalability, and flexibility, such as the SaaS or PaaS models. They are great for client businesses that do not wish to have an in-house IT department or infrastructure costs. However, the usage of Cloud means two important things: service costs (especially when scaling up and adding features), and a company's retention of data ownership, which means security issues.

5. **Open-Source Software:** This is open and *free* source code that can be freely downloaded, used, shared, and modified, or distributed, and often used by Linux solutions. However, recently, Microsoft has said they are going to open-source software. This usually does come with limitations of the free *license*, especially from Apache. Typically, this is the opposite of closed-source code, such as Apple IOS. Some of the downsides to Open-source code are not being profitable, still requiring a license, and may include boilerplate or bad code. With this type of software, the source code is publicly available, allowing users to modify and distribute it. While it can be a low-cost option, it might require technical expertise for setup and support.
6. **In-House Development:** This means building a software system from the ground up using your own development team. This method gives you total control over the final product and its features, ensuring it is tailored to your specific business needs.

DealAura System Acquisitions

The *DealAura* software system will be developed in-house and deployed with cloud computing.

- **In-House Development:** This software system proposal will be developed by using proprietary code, methods, and in-house building and source coding. This approach allows for the program to be a highly secure, customized solution with AI and BI features that are unique and proprietary. This software will also come with a software license.
- **Cloud Computing:** *DealAura* is a cloud-based software application that is primarily used by *deploying* it through a cloud service to either desktop or mobile devices. The usage of cloud computing for deploying the software program allows clients or customers to be able to access it from anywhere, at any time, provided they have internet access available. The cloud models to be used for deployment are SaaS and PaaS. The cloud service provider options include AWS, Azure, and Google Cloud.

Project Plan: Gantt Chart

Task ID	Task Name	Start Date	End Date	Duration	Predecessor
1 Phase 1: Waterfall Requirements					
1.1	Requirements Gathering	Sept 1, 2025	Sept 30, 2025	30 days	N/A
1.2	System Design & Prototyping	Oct 1, 2025	Oct 31, 2025	31 days	1.1
1.3	Documentation and Review	Oct 31, 2025	Nov 1, 2025	1 day	1.2
2 Phase 2: Agile Sprints					
2.1	Sprint 1: Core Feaures	Nov 1, 2025	Nov 15, 2025	15 days	1.3
2.2	Sprint 2: AI and added features	Nov 16, 2025	Nov 30, 2025	15 days	2.1
2.3	Sprint 3: Final Build feaures	Dec 1, 2025	Dec 15, 2025	15 days	2.2
2.4	Sprint 4: API Integration for Desktop/Mobile	Dec 16, 2025	Dec 30, 2025	15 days	2.3
3 Phase 3: Testing & Deployment					
3.1	System and User Acceptance Testing (UAT)	Jan 1, 2026	Jan 15, 2026	15 days	2.4
3.2	Cloud Deployment & Release	Jan 16, 2026	Jan 20, 2026	5 days	3.1
4 Phase 4: Post-Launch and Support					
4.1	Ongoing Maintenance and Updates	Jan 21, 2026	Indefinite	Indefinite	3.2

Figure 1.1 Gantt Chart

Feasibility Analysis

1. Economic Feasibility Analysis

The economic feasibility of the *DealAura* program and design rests on balancing the costs and benefits to the proposal. While the estimated and calculated costs will be high, the reward or benefits will also be equally high or greater than the costs. The following benefits and costs are estimates that are based on general practice and software development industry standards.

a. Tangible Benefits

- Increased profitability: By offering this solution and program to businesses, they will directly be able to benefit from sales and profitability, which in turn will increase the popularity, sales, and profitability of *DealAura* program and its developers.
- Reduced overhead and provides cost savings: The program will be able to provide a reduction in overhead and other costs to business, which in turn makes the program more valuable to business for cost savings.
- Environmentally Friendly: The program is environmentally friendly, does not cause pollution to the environment or company's environment costs.
- Highly Secure Program: Another benefit to this program is that it is also highly secure, which helps business reduce their vulnerable and security risks, reducing such cybersecurity costs.

- Highly scalable in cloud: This program is highly scalability for enterprise businesses who use cloud deployments, which in turn helps reduce cloud service level costs.

b. Costs (One-Time)

- Software Development Staff: As an in-house development, a significant cost is for developers, designers, architects, and managers. The minimum of \$200,000 - \$350,000 or more for a team of 3-6, and for a 6–12-month development phase.
- Cloud Infrastructure Setup: Cloud infrastructure setup costs on a service like AWS, Azure, or Google Cloud. \$5,000 - \$20,000.
- Licensing and Tools: Development licenses and software tools costs \$2,000 - \$5,000.
- Testing and QA: Costs for quality control and assurance, or bench testing, \$10,000 - \$30,000.
- Marketing and Rollout: Cost to launch the product onto the market is \$15,000 - \$50,000.

c. Costs (Recurring)

- Other Staff Salaries: Other staffing and ongoing staff for maintenance is yearly recurring cost of \$150,000 - \$250,000.
- Cloud Computing Services: Costs for any cloud usage (SaaS and PaaS models) based on scaling and data storage, \$2,000 - \$10,000 per month.
- Licensing and Support: Recurring costs for software licenses, third-party APIs or tools, other support agreements, or contracts, \$500 - \$2,000 per month.
- Marketing and Customer Service: Costs for marketing, sales, advertising, campaigning, and customer service, \$5,000 - \$15,000 per month.

d. Intangible Benefits and Costs

- Benefits: The program offers a strong *competitive advantage* for businesses, by consolidating sales functions and automating time-intensive processes. It can also improve sales and team overall satisfaction by reducing such dull tasks and allowing them to focus on high-level strategy, planning, and professional relationships.
- Costs: Intangible costs include the training and learning curve for salespersons and teams to use and adopt the new program, and any minimal risk of the program failing to meet business needs or being overpriced or under budgeted by the businesses.

Category	Item	Type	Estimated Cost/Benefit	Notes
Tangible Benefits	Increased Revenue	Benefit	Variable (High)	The system is designed to increase profitability by streamlining sales processes.
	Reduced Overhead Costs	Benefit	Variable (High)	The system helps reduce business and salesforce overhead.
One-Time Costs	Software Development St	Cost	\$200,000 - \$350,000+	Includes salaries for in-house developers, designers, and project managers.
	Initial Infrastructure Setup	Cost	\$5,000 - \$20,000	For cloud services like AWS, Azure, or Google Cloud.
	Licensing and Tools	Cost	\$2,000 - \$5,000	For development software and tools.
	Initial Testing and QA	Cost	\$10,000 - \$30,000	Costs for security and functionality testing.
Recurring Costs	Marketing and Rollout	Cost	\$15,000 - \$50,000	Initial costs to launch the product to market.
	Staff Salaries	Cost	\$150,000 - \$250,000+ (annually)	Ongoing salaries for maintenance and future development.
	Cloud Computing Services	Cost	\$2,000 - \$10,000+ (monthly)	Based on scalability and usage.
	Licensing and Support	Cost	\$500 - \$2,000+ (monthly)	For third-party software and API access.
Intangible Benefit	Competitive Advantage	Benefit	High	The system provides a significant competitive advantage by consolidating sales functions.
	Improved Employee Morale	Benefit	High	Reduces manual, repetitive tasks, allowing salespeople to focus on strategic activities.
Intangible Costs	Increased Security	Benefit	High	The in-house development approach allows for a highly secure solution.
	Learning Curve	Cost	Variable	Time and effort required for staff to adopt the new system.
	Project Failure Risk	Cost	High	The risk that the project may not meet its goals or may exceed its budget.

Figure 2.1 Economic Feasibility Analysis in Excel file.

2. Technical Feasibility Analysis

The technical feasibility of the *DealAura* program to be successful is extremely high, as the proposed solution practices and employs well-established methods and procedures for software development and cloud deployment. The program's technical risks are well within manageable range, provided proper management and maintenance oversight. The following are technical risk factors:

- **In-House Development:** Building the program with a proprietary source code gives total control over the final product. Project success rests on the in-house team, and does not have to depend on outside variables, factors, or risks.
- **AI and LLM Integration:** The core of *DealAura* program using AI/ML and LLMs for tasks which in a built-in core component to the program integration. Success here depends on how well the AI models are set up, written, trained, and functioning.
- **Backend API Integration:** The program is designed to be able to connect or integrate with existing CRM platforms through microservices or backend API. This requires vigilant architectural and development planning, design, and setup, to ensure seamless and secure connections between different systems.
- **Security:** The program and its source code are highly secure. This follows the *DevSecOps* models, or security first and not last, in the software build. Security is a constant element to be monitored during the SDLC. To ensure these risks are reduced or met, software architect and cyber security professionals will also be employed. Also to mitigate risks, businesses understand they own the data risk while operating the program on cloud.
- **Scalability:** The cloud-based program solution is chosen for its scalability and flexibility. While nominal usage of the program in cloud is not likely to incur heavy cloud service

costs, cloud scaling up or out can lead to increased service costs, depending on the business and its cloud service provider.

- **Overall Technical Risk Rating: Moderate level.** The program is highly productive, operational, secure, and reliable. However, there are known and inherited risks involved. These risks are mitigated or offset by the proactive steps, policies and practices implemented by the developers' team.

3. Operational Feasibility Analysis

DealAura program is also highly operationally feasible. The proposed program software is well made, runs well, solid build, very secure, highly scalable, and user friendly. There are little issues within the operations of the program, or its usage. Likewise, the program is very operational and affordable for businesses and does address and solve many business and sales teams' problems.

a) Solving Business and Sales Problems

The program is designed to solve the significant business challenges of sales teams, who spend much time on highly manual and redundant clerical or office requirements. By automating these tasks like prospecting, drafting letters, writing, or sending proposals, *DealAura* solves a major inefficiency issue and reduces a sales team's time and cost for revenue-generating activities.

b) Changes in Day-to-Day Business Activities

Salespeople and Sales Team: Salespersons and their teams will be able to reallocate their time and costs to other areas which are more productive for human activity. This program will allow for much of the hard manual labor to be automated and AI used, such that sales and their revenues will come much faster, easier, and more efficiently. Sales teams can also rest assured that they are getting more value and proven productivity in this program, at reduced cost, versus the traditional approach.

Managers: The program will enable sales or other business managers to gain great insights into their sales figures, forecasting, revenues, etc., by pulling or receiving AI generated BI reports. They will know how to better deal with business issues and optimize the company's goals.

4. Schedule, Legal & Contractual, and Political Feasibility Analysis

a. Schedule Feasibility

The proposed program's schedule is feasible due to its use of a *hybrid* development model. The initial phase or *kickoff* uses the traditional Waterfall approach for the requirements and design, which ensures a SOLID foundation and clear project and product charter plans. This will include the SRD. Then comes the transition to the Agile method, which is for the rest of the software and product lifecycle. Agile provides schedule flexibility and slack tolerance to handle ongoing issues and deploy quick releases to market. This blend of methodologies helps to mitigate the risk of a rigid schedule or timeline issue.

b. Legal & Contractual Feasibility

The program and its development follow all applicable laws and requirements. The program will require legal and contractual obligations, including the terms of software licensing. The usage of backend API for integration with other applications will require some contractual obligations. Developers make clear to businesses that they own the data, and that the program nor its developers are liable for any data storage or data usage, especially in cloud. The developers may provide maintenance contract to continue to oversee the program's maintenance as to ensure any updates or new releases are made successfully. The program is highly secure, but the developers also may include an additional security contract offer or insurance as to oversee the security of the program while at REST or while in usage.

c. Political Feasibility

The program has a high degree of political feasibility. First, due to the nature of the program, businesses and sales teams will strongly favor the program and its usage. The program is likely to receive positive feedback from end-users, customers, clients, surveys.. Thus, there should be no foreseeable issues within political spectrum. The program, and its AI models, are politically agnostic, such that anyone should be able to benefit, enjoy the usage without issue.

Requirements Determination

Requirements Determination Processes/Methods

To ensure the DealAura software system is a secure, user-friendly, and highly effective solution, a combination of requirements determination processes will be employed. The project's hybrid methodology, which begins with a Waterfall approach, necessitates a thorough and well-

documented requirements gathering phase. The primary methods used will be Interviews, Observations, and Documentation Analysis. While other methods such as surveys and Joint Application Design (JAD) sessions are valuable, they are not the primary focus for this phase. Surveys can be too general for the specific, proprietary nature of this in-house software, and JAD sessions are more beneficial during the design phase. For this initial phase, a more focused, one-on-one approach preferred to capture qualitative data and deeply understand user pain points.

Interviews

Interviews will be a critical part of the requirements gathering process. They will allow for direct, detailed, and qualitative feedback from key stakeholders.

1. Audience: Sales Professionals
 - a) Purpose: To understand their current daily workflows, pain points, and specific needs for an automated sales tool.
 - b) Sample Questions:
 1. Describe your typical day. What manual tasks do you perform regularly?
 2. What are the most time-consuming or repetitive parts of your job?
 3. What information do you need most often to close a deal?
 4. How do you currently track your leads and progress?
 5. What features would make your job easier and more productive?
2. Audience: Sales Managers
 - a) Purpose: To determine their needs for reporting, business intelligence, and team management tools.
 - b) Sample Questions:
 1. What key performance indicators (KPIs) or metrics do you currently track your team?
 2. What kind of reports would provide you with the most valuable insights into team performance?
 3. How would you use AI-generated insights to coach your team or make strategic decisions?
 4. What are the biggest challenges you face in managing your team's sales pipeline?

3. Audience: Information Technology (IT) Staff

- a) Purpose: To define the technical, security, and integration requirements.
- b) Sample Questions:
 - 1. What existing CRM platforms or other systems would DealAura need to integrate with?
 - 2. What are the company's current security protocols and data privacy policies?
 - 3. What are the technical specifications for cloud deployment?
 - 4. Are there any specific API or data access limitations we should be aware of?

Observations

Observing sales team members and managers will provide valuable insights into their "as-is" processes, which may not be fully captured during interviews.

- 1. Who will be observed: Sales team managers and members.
- 2. What will be observed:
 - a) How sales professionals navigate their current software and tools.
 - b) Which tasks are most prone to error or manual entry?
 - c) The time it takes to complete a specific task.
 - d) Any workarounds or unofficial processes they have developed to streamline workflow.

Documentation Analysis

Analyzing existing documentation will provide a foundational understanding of the current business environment and technical landscape.

- 1. What documentation will be collected:
 - a) Existing sales process manuals and training materials.
 - b) Sample sales reports and performance dashboards.
 - c) Technical documentation for existing CRM systems or other integrated software.
 - d) Job descriptions for sales professionals and managers.
 - e) Company policies on data security and privacy.

2. How this documentation will assist:
 - a) It will provide a baseline understanding of the current system and processes.
 - b) It will help identify formal business rules, data formats, and reporting requirements.
 - c) It will confirm or challenge the information gathered during interviews and observations.
3. What the team will be looking for:
 - a) Discrepancies between documented processes and actual observed behavior.
 - b) Specific data fields or metrics that are critical to the business.
 - c) Constraints or limitations of existing systems that DealAura must overcome.
 - d) A comprehensive list of terms and definitions used within the sales department.

Data Flow Diagram (DFD) for DealAura

This DFD shows how DealAura takes data from end users and external systems, processes it with AI models, data stores, or lakes it, and then presents it in a useful way to managers.

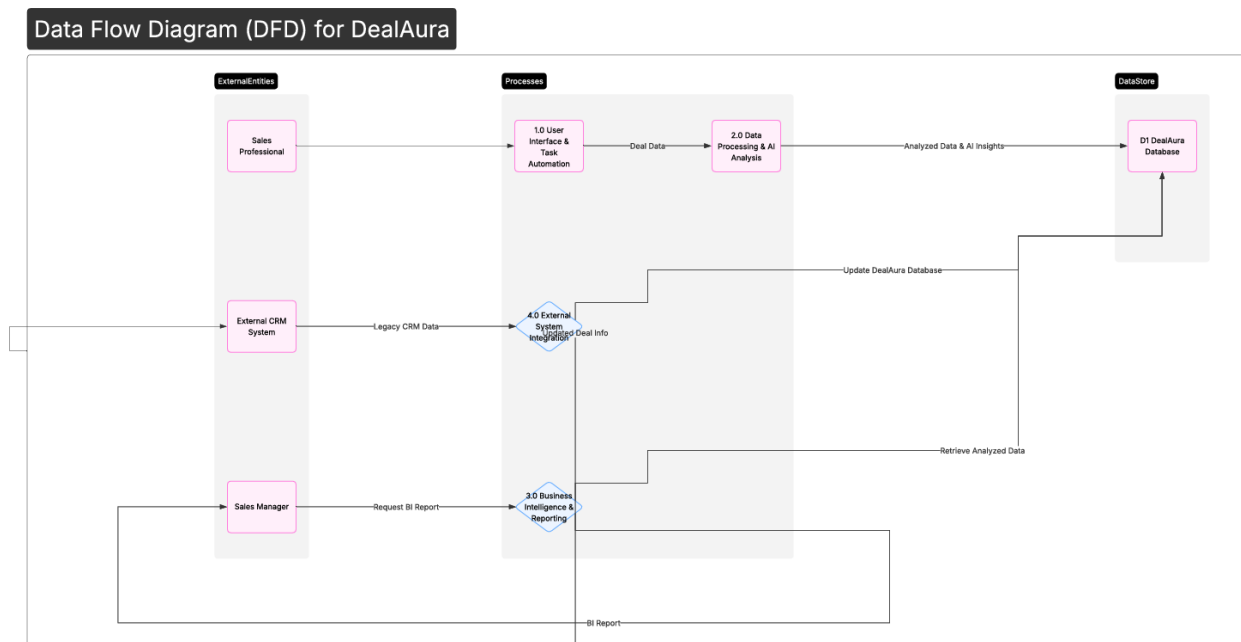


Figure 3.1 DFD Chart for DealAura

External Entities

- Sales Professional: The primary end-user who interacts with the system to perform daily sales functions or tasks.

- Sales Manager: The end user who needs to access BI reports and oversee sales team performance and metrics.
- External CRM System: The existing CRM platform that DealAura may integrate with.

Processes

- 1.0 User Interface & Task Automation: This process represents the user-facing part of the application. It manages all user input from the Sales Professional and automates a variety of tasks.
- 2.0 Data Processing & AI Analysis: This is the core engine of the DealAura system. It receives raw data and processes it.
- 3.0 Business Intelligence (BI) & Reporting: This process takes the analyzed data and presents it in a comprehensible format. It generates reports, dashboards, and visualizations for the Sales Manager.
- 4.0 External System Integration: This process manages communication with external systems, ensuring data can be imported from and exported to other platforms like a company's main CRM.

Data Stores/ Data Lakes / Data Warehouse (stored in Cloud)

- D1 DealAura Database: This is the central repository for all data within the system. It stores raw deal information, user profiles, AI-generated insights, and historical data.

Data Flows

- Flow 1: The Sales Professional enters the user data into the 1.0 User Interface & Task Automation process.
- Flow 2: The 1.0 User Interface & Task Automation process sends raw data to the 2.0 Data Processing & AI Analysis process.
- Flow 3: The 1.0 User Interface & Task Automation process sends Automated Task Requests to the 2.0 Data Processing & AI Analysis process.
- Flow 4: The 2.0 Data Processing & AI Analysis process generates and sends Analyzed Data & AI Insights to the D1 DealAura Database for storage.
- Flow 5: The Sales Manager requests BI Report from the 3.0 Business Intelligence & Reporting process.
- Flow 6: The 3.0 Business Intelligence & Reporting process retrieves Analyzed Data from the D1 DealAura Database to compile the report.

- Flow 7: The 3.0 Business Intelligence & Reporting process sends BI Report to the Sales Manager.
- Flow 8: The External CRM System sends Legacy CRM Data to the 4.0 External System Integration process.
- Flow 9: The 4.0 External System Integration process updates the D1 DealAura Database with Synchronized Data.
- Flow 10: The 4.0 External System Integration process exports Updated Deal Info back to the External CRM System.

Dialogue Diagram

A dialogue diagram maps out the user's path going through the system. It shows the sequence of screens, the user actions that triggers the transitions, and the system responses. Our diagram will be based on a typical workflow for a sales professional using DealAura.

Key Screens (Interface):

- S-1 (Main Dashboard): The first screen a user sees upon login. It provides a high-level overview.
- S-2 (Deals List): Displays a list of all deals, searchable and filterable.
- S-3 (Deal Details): Shows all information related to a specific deal.
- S-4 (New Deal Form): A form for creating a new deal.
- S-5 (Reports Menu): A list of available reports.
- S-6 (Specific Report View): Displays the data and charts for a chosen report.

User Actions & System Responses:

1. From S-1 (Dashboard):
 - User Action: Click "View All Deals."
 - System Response: Navigate to S-2 (Deals List).
 - User Action: Click "View Report."
 - System Response: Navigate to S-5 (Reports Menu).
2. From S-2 (Deals List):
 - User Action: Click a specific deal from the list.
 - System Response: Navigate to S-3 (Deal Details).

- User Action: Click "New Deal" button.
 - System Response: Navigate to S-4 (New Deal Form).
3. From S-3 (Deal Details):
- User Action: Click "Save" after editing.
 - System Response: Update the database and return to S-2 (Deals List).
 - User Action: Click "Return to List."
 - System Response: Navigate back to S-2 (Deals List).
4. From S-4 (New Deal Form):
- User Action: Submit the completed form.
 - System Response: Create a new record in the database and navigate to S-3 (Deal Details) for the newly created deal.
5. From S-5 (Reports Menu):
- User Action: Select a specific report (e.g., "Performance Report").
 - System Response: Generate the data and display it in S-6 (Specific Report View).
6. From S-6 (Specific Report View):
- User Action: Click "Back to Reports."
 - System Response: Navigate to S-5 (Reports Menu).

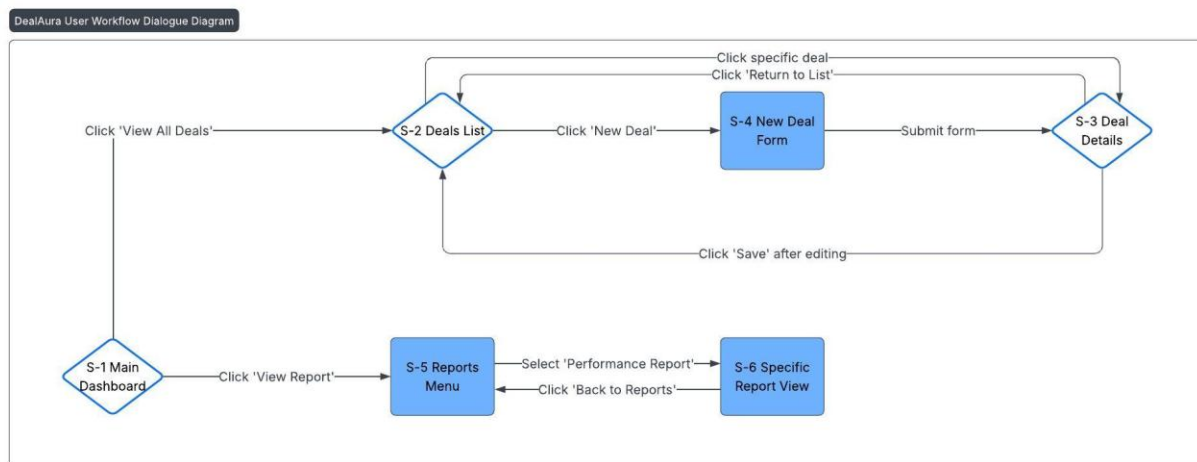


Figure 4.1 DealAura Dialogue Diagram1

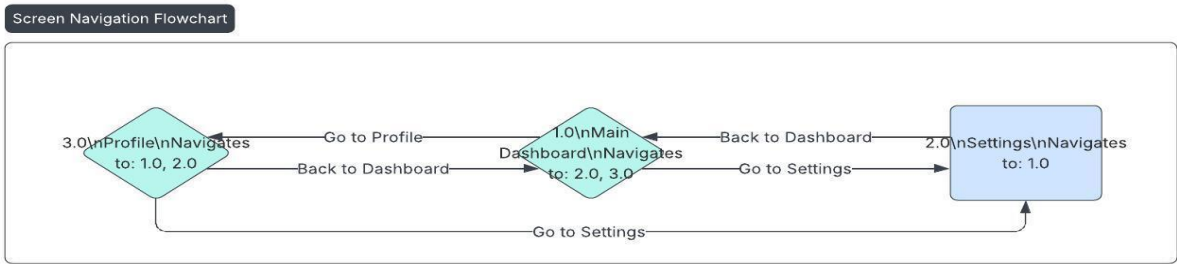


Figure 5.1 DealAura Dialogue Diagram 2

ER Diagram

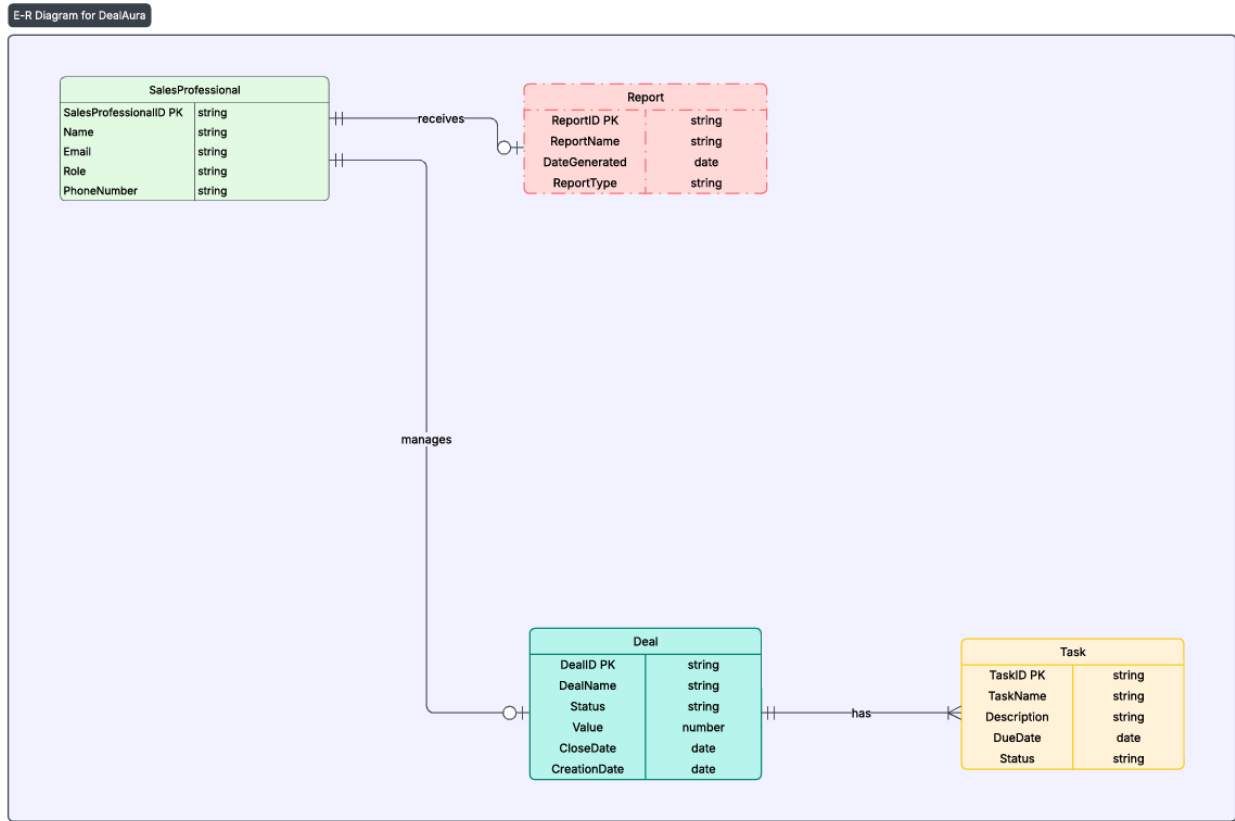


Figure 6.1 ER Diagram.

System Wireframe (LoFi)

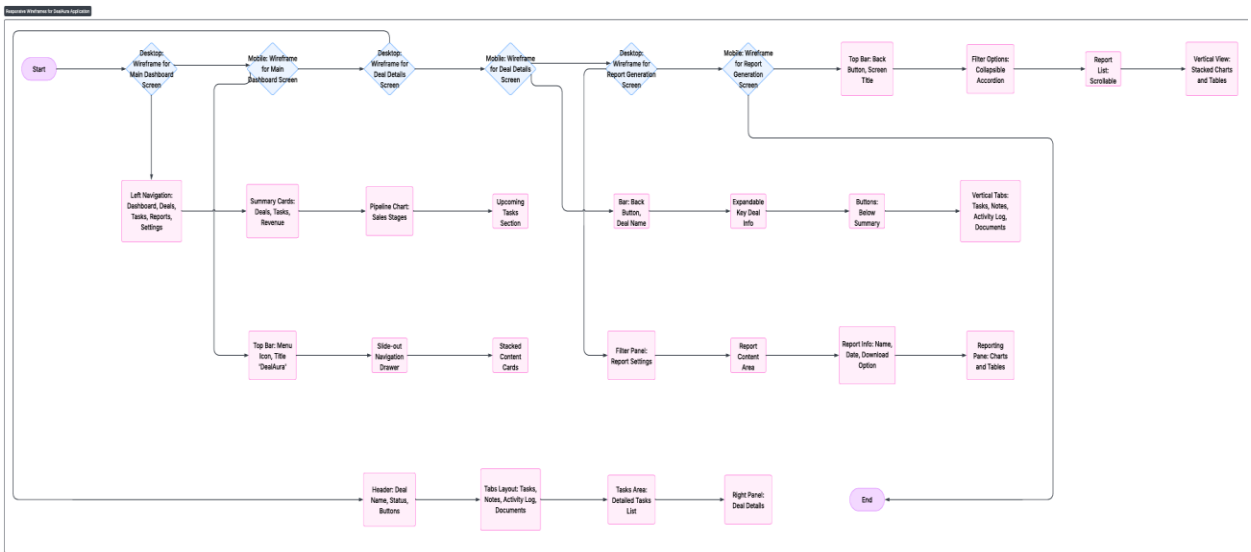


Figure 7.1 System Wireframe

HiFi Wireframes

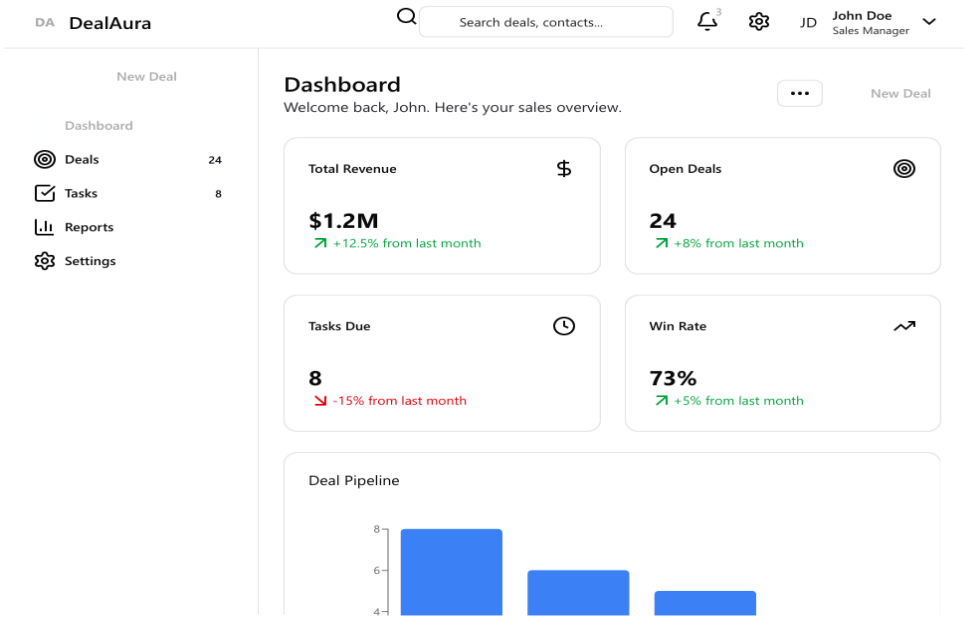
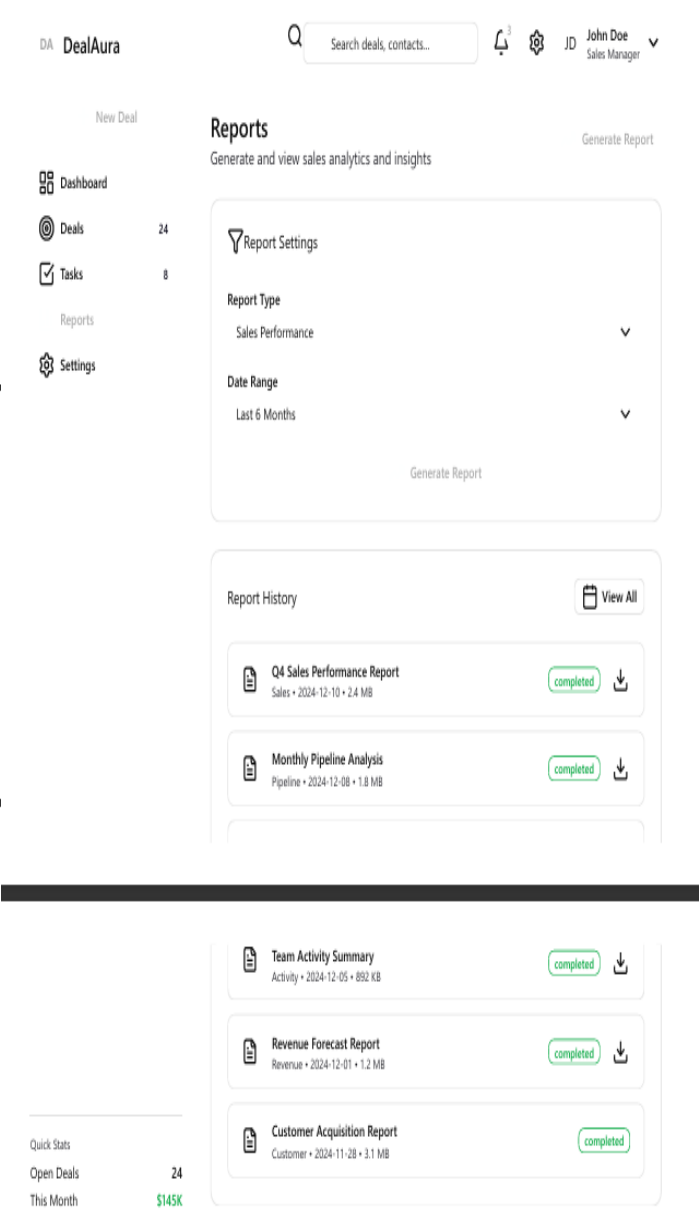
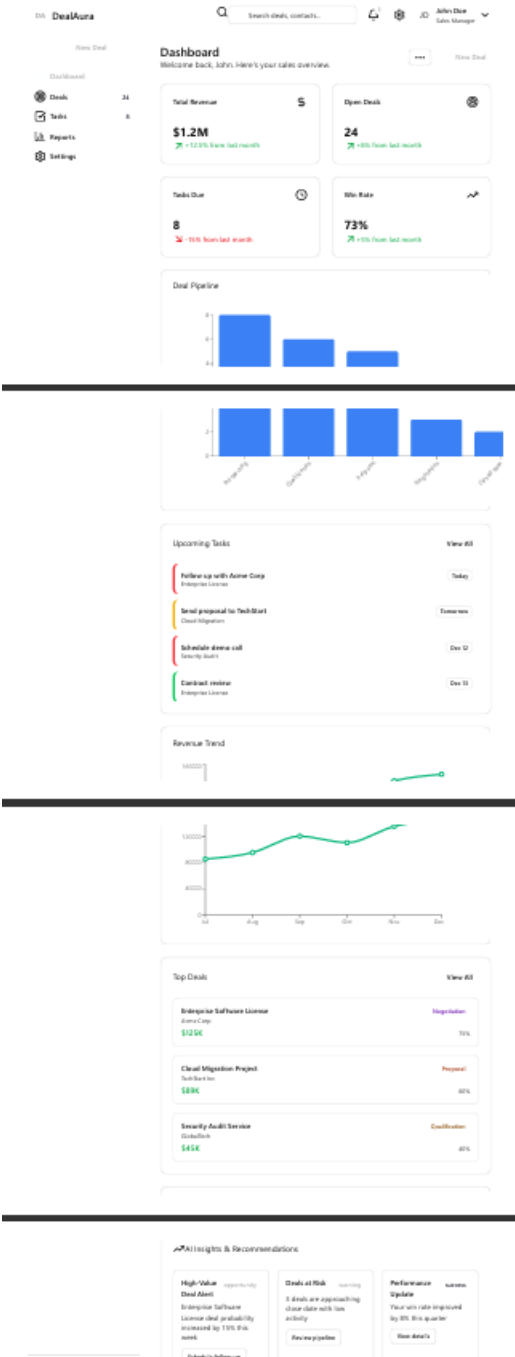


Figure 8.1 HiFi Wireframe - Dashboard/ Main Menu screen



Figures 9.1 and 10.1 HiFi wireframes

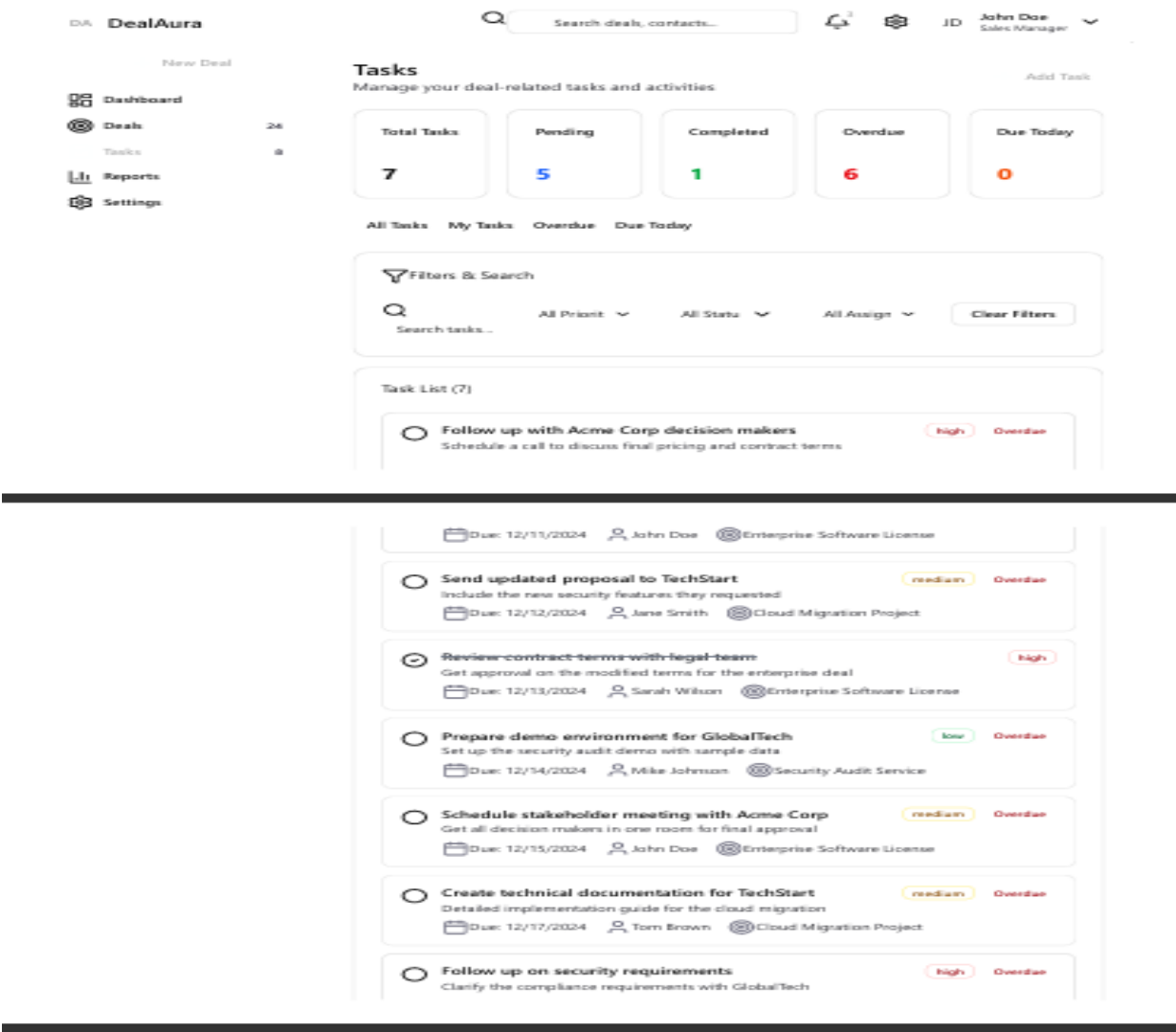


Figure 11.1 HiFi wireframes

Implementation Plan

The implementation plan details the activities required to transition from a design to a fully operational system. For DealAura, this involves a combination of technical and non-technical tasks.

1. **Coding Plan:** The development will follow the hybrid methodology outlined in the Gantt chart. We will start with the Waterfall approach for the initial infrastructure and core

components, then transition to Agile sprints for feature development. The focus will be on modular, reusable code to enable rapid deployment and easy maintenance.

2. **Testing Plan:** A multi-phase testing plan will be implemented. This includes:
 - Unit Testing: Individual code modules will be tested by developers.
 - Integration Testing: Different modules will be tested together to ensure they work seamlessly.
 - System Testing: The complete system will be tested to ensure it meets all functional and non-functional requirements.
 - User Acceptance Testing (UAT): A group of sales professionals will test the application to ensure it is user-friendly and meets their needs.
3. **Installation Plan:** Since DealAura is a cloud-based SaaS, the installation will primarily involve deploying the application to a cloud service provider (AWS, Azure, or Google Cloud). It will be made available via a secure URL for desktop access, and through app stores for mobile access. There will be no on-premise installation required for the client.
4. **Documentation Plan:** The documentation will be created for both technical and end users.
 - a. Technical Documentation: This will include detailed API documentation, code comments, and system architecture guides for future development and maintenance.
 - b. User Documentation: This will include a user manual, a frequently asked questions (FAQ) section, and video tutorials to guide inexperienced users through the system.
5. **Training Plan:** The training will be tailored to the user's role.
 - a. Sales Professionals: Will receive hands-on training on how to use the automation features and the mobile application.
 - b. Sales Managers: Will be trained in how to interpret and use the BI reports and manage their teams within the application.
6. **Support Methods:** A multi-tiered support system will be established.
 - a. Tier 1: An online help desk and a dedicated email support channel for basic inquiries.
 - b. Tier 2: A team of technical specialists for more complex issues.
 - c. Tier 3: Direct access to a lead developer for critical bug fixes.

Maintenance Plan

A maintenance plan ensures that the software remains functional, secure, and up to date. For DealAura, a proactive approach to maintenance is essential for long-term success.

1. **Corrective Maintenance:** This involves fixing defects or errors that are discovered after the system is launched.
 - **Implementation:** A bug-tracking system will be used to log and prioritize issues reported by users. A dedicated team will be assigned to address critical bugs as a top priority, with a goal of releasing fixes within a specified timeframe.
2. **Adaptive Maintenance:** This involves adapting the system to changes in the environment, such as the latest version of an operating system or a change in API requirements.
 - a. **Implementation:** The team will actively monitor external services and technologies (cloud provider updates, changes to third-party APIs). Development sprints will be allocated to implementing necessary changes to ensure the application remains compatible and functional.
3. **Perfective Maintenance:** This involves improving the system's performance, usability, or features based on user feedback.
 - a. **Implementation:** We will collect user feedback through surveys and a feedback portal within the application. This feedback will be analyzed to identify common pain points or popular feature requests. These will be prioritized and incorporated into future agile development sprints, ensuring the product continuously improves.
4. **Preventive Maintenance:** This involves making changes to the system to prevent future problems or to improve its maintainability.
 - a. **Implementation:** Regular code reviews, security audits, and server monitoring will be conducted to identify potential issues before they become critical. This includes optimizing database queries, refactoring code to improve readability, and updating internal libraries to their latest versions.

Product Requirements Document (PRD): Deal Aura

1. Executive Summary

- **Product Name:** Deal Aura.
- **Purpose:** A cloud-based sales tool featuring AI and Business Intelligence (BI) capabilities designed to automate manual, repetitive clerical tasks for sales teams.
- **Background:** Modern sales teams lose significant time to low-productivity tasks like manual prospecting, drafting outreach, and creating proposals. Deal Aura consolidates these functions into one platform to increase revenue-generating activities.

2. Goals

- **Streamline Workflow:** Automate searching, booking, and planning functions.
- **Increase Productivity:** Enable sales professionals to focus on strategic planning and relationship building.
- **Ensure Accuracy & Security:** Provide a highly secure, proprietary system with deep-learning AI.
- **Reduce Overhead:** Lower business costs through efficiency and automation.

3. Stakeholders

- **Sales Professionals:** Primary end-users utilizing automation for daily tasks.
- **Sales Managers:** Secondary users accessing BI reports for team performance tracking and forecasting.
- **IT Staff:** Responsible for technical specifications, security, and cloud deployment.
- **Software Development Team:** In-house developers, designers, and architects responsible for building and maintaining the system.

4. Problem and Pain Points

- **Manual Redundancy:** High volume of repetitive clerical requirements.
- **Time Inefficiency:** Significant hours devoted to manual prospecting, research, and personalized outreach.
- **Profit Loss:** Inefficiencies directly reduce revenue-generating activities and business strategy effectiveness.

5. Requirements

- **Functional Requirements:**

- **Personalized Sales Research:** Deep-level research into prospects and demographics.
- **Pitch & Proposal Generator:** AI-drafted personalized pitches, pitch decks, and sales agreements.
- **Threat Detector:** KPI tracking to identify negative feedback and potential contract cancellations.
- **BI Reporting:** Generation of dashboards and visualizations for managers.
- **Non-Functional Requirements:**
 - **Security:** Dev Sec Ops model with secure source code.
 - **Scalability:** Cloud-based architecture (SaaS/PaaS) designed to scale with enterprise needs.
 - **Integration:** Backend API for seamless connection with existing CRM platforms.

6. User Stories

- **As a Sales Professional,** I want to automate my daily prospect research so I can spend more time in meetings.
- **As a Sales Manager,** I want to receive AI-generated BI reports so I can gain insights into sales figures and forecasting.
- **As an IT Administrator,** I want a secure API to integrate Deal Aura with our current CRM system without data leakage.

Product Roadmap: Deal Aura Development Lifecycle

The project follows a Hybrid Methodology, utilizing Waterfall for initial design and documentation, then transitioning to Agile for development and maintenance.

Phase	Timeline	Focus Area	Key Activities
Phase 1: Foundation (Waterfall)	Month 1-2	Requirements & Design	Documentation, SRD creation, and system architecture planning

Phase	Timeline	Focus Area	Key Activities
Phase 2: Core Build (Agile Sprint 1)	Month 3-5	Infrastructure & AI Core	Setting up cloud infrastructure (AWS/Azure/GCP) and AI/LLM model training
Phase 3: Integration (Agile Sprint 2)	Month 6-8	API & Features	Developing backend APIs for CRM integration and primary automation features
Phase 4: QA & UAT (Agile Sprint 3)	Month 9-10	Testing & Quality	Unit testing, integration testing, and User Acceptance Testing with sales pros
Phase 5: Launch & Support	Month 11-12	Deployment & Training	Cloud rollout, training sessions for staff/managers, and Tier 1 support setup
Post-Launch: Growth	Ongoing	Maintenance	Proactive bug tracking (Corrective), API updates (Adaptive), and new features (Perfective)